

ARCOS Group

uc3m | Universidad **Carlos III** de Madrid

L3: Fundamentals of assembler programming (4) **Computer Structure**

Bachelor in Computer Science and Engineering
Bachelor in Applied Mathematics and Computing
Dual Bachelor in Computer Science and Engineering and Business Administration



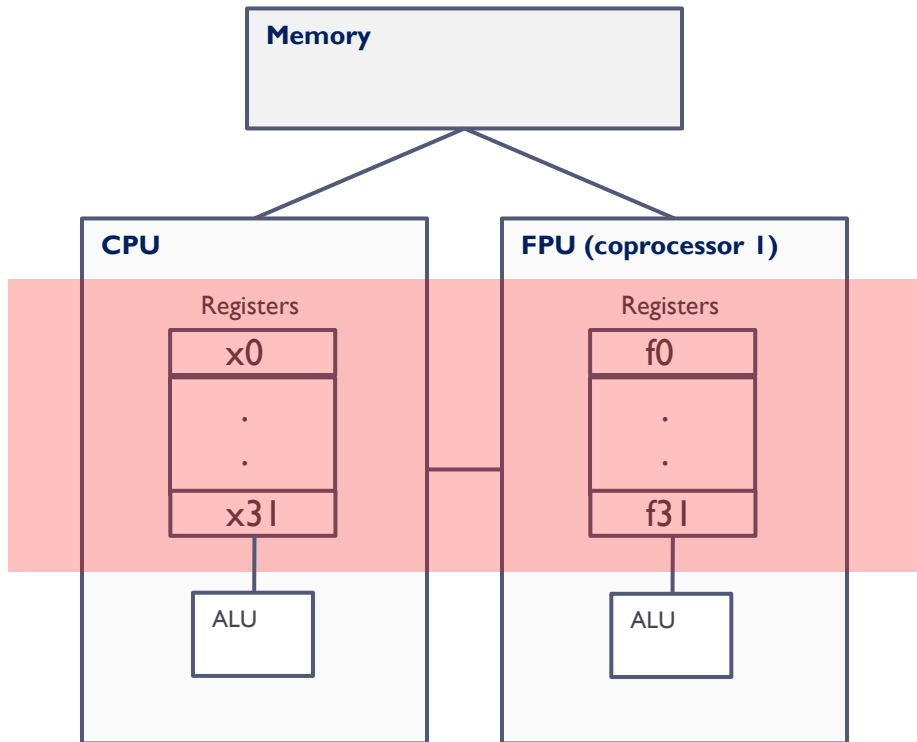
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1. Basic concepts on assembly programming
2. RISC-V 32 assembly, memory model and data representation
3. Instruction formats, addressing modes, and instruction sets
4. **Floating point in assembly**
 - ▶ Registers
 - ▶ Data transfer
 - ▶ Arithmetic operations
 - ▶ Detect and compare
 - ▶ Memory
5. Procedure calls and stack convention

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RISC-V 32



▶ Registers

- ▶ Inside the processor
- ▶ Instructions work with values in registers
- ▶ 32 registers of 32 bits

▶ Memory

- ▶ Outside the processor
- ▶ Data exchange to/from registers
- ▶ More capacity but longer access time than registers
- ▶ 32-bit addresses
 - ▶ 4 GiB addressable

Register file (integer)

summary

Symbolic name	Number	Usage
zero	x0	Constant 0
ra	x1	Return address (used by function calls)
sp	x2	Stack pointer
gp	x3	Pointer to global area
tp	x4	Thread pointer
t0...t2	x5-x7	Temporary (NO preserved across calls)
s0/fp	x8	Temporary (preserved across calls) / Frame pointer
s1	x9	Temporary (preserved across calls)
a0...a1	x10...11	Input function arguments / return values
a2...a7	12...x17	Input function arguments
s2... s11	x18...x27	Temporary (preserved across calls)
t3...t6	x28...x31	Temporary (NO preserved across calls)

- ▶ There are 32 registers
 - ▶ Size: 4 bytes (1 word)
 - ▶ Dual name: logical and numerical (starts with **x** at the beginning)
- ▶ Use convention
 - ▶ Reserved
 - ▶ Arguments
 - ▶ Results
 - ▶ Temporary
 - ▶ Pointers

Register File (floating-point)

ABI Name	Name	Usage
ft0...ft7	f0 ... f7	Temporals (like t...)
fs0...fs1	f8 ... f9	There are saved (like s...)
fa0...fa1	f10 ... f11	Arguments/return (like a...)
fa2...fa7	f12 ... f17	Arguments (like a...)
fs2...fs11	f18 ... f27	There are saved (like s...)
ft8...ft11	f28 ... f31	Temporals (like t...)

- ▶ There are 32 registers
- ▶ The ft0 register does not have its value set to 0.
- ▶ In the single precision extension, the registers are 32 bits (4 bytes).
- ▶ In the double precision extension, the registers are 64 bits (8 bytes) and can store:
 - ▶ Single precision values in the lower 32 bits of the register
 - ▶ Double precision values in the 64 bits of the register

Copy instructions

(floating-point registers $\leftrightarrow$ floating-point registers)

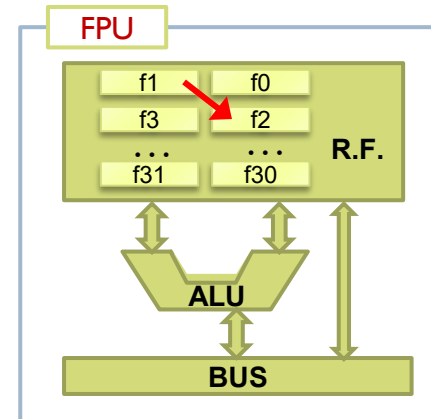
- ▶ Copy data:

- ▶ Between FPU registers
- ▶ Between CPU registers and FPU registers

- ▶ Examples:

- ▶ Copy register to register (.s .d):

`fmv.s f2, f1` *# rd = rs*

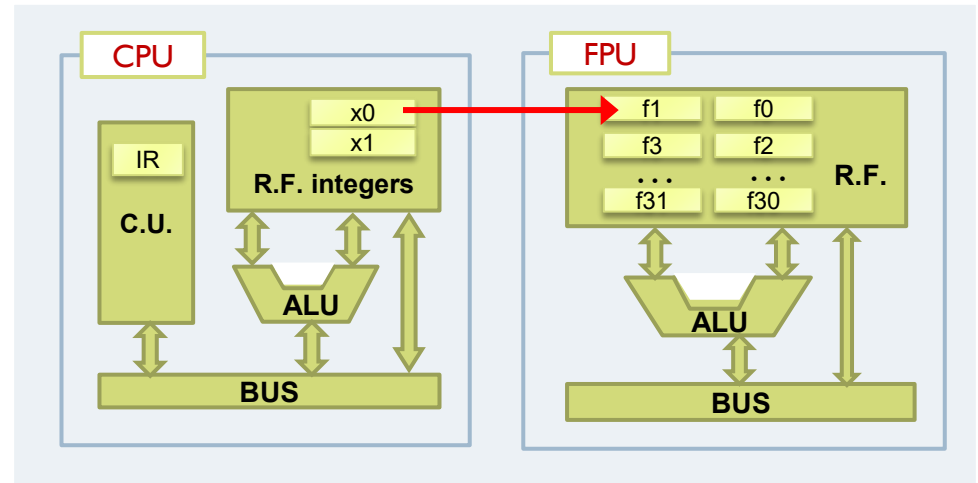


Copy instructions

(integer registers $\leftrightarrow$ floating-point registers)

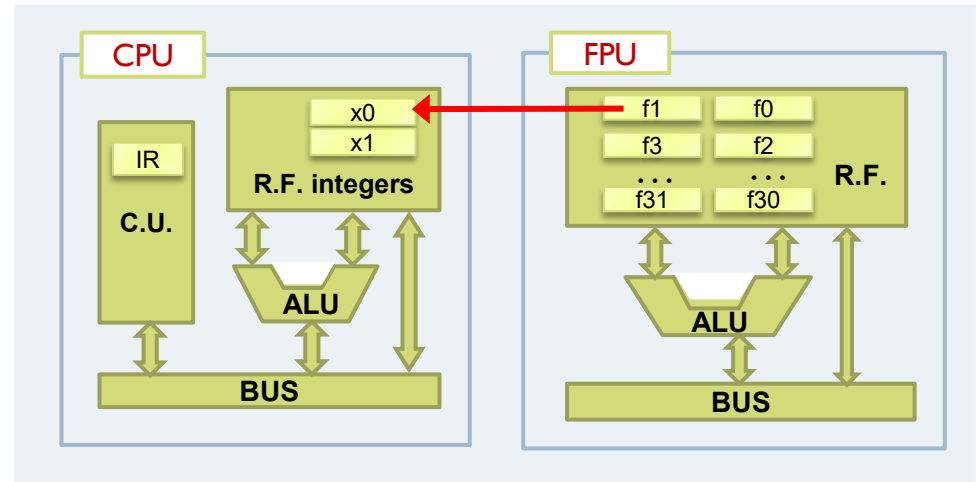
`fmv.w.x rds rsw`

- Copy from integer reg. `rs` to a floating register `rd` (*single precision*)



`fmv.x.w rdw rss`

- Copy from floating register `rs` (*single precision*) to an integer register `rd`



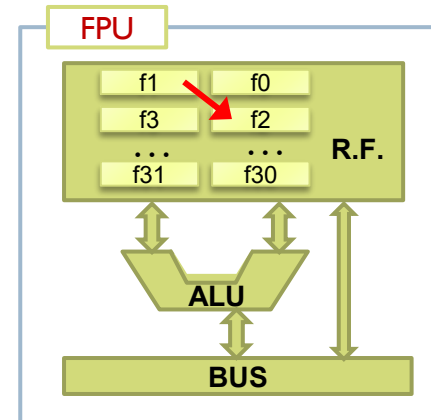
Copy instructions

(floating-point registers $\leftrightarrow$ floating-point registers)

- ▶ It does not exist `fli.s` and `fli.d` (*):
 - ▶ The `Zfa` extension provides `fli.s` and `fli.d` ...
 - ▶ ... But these instructions use a 5-bit index to immediately load one of the 32 most common constant values in scientific and mathematical software (1.0, 0.5, min, inf, etc.)

- ▶ Examples:

- ▶ ~~Immediate load~~
`fli.s f0, 5.0`



Example

```
float PI = 3.1415;
```

```
.text
```

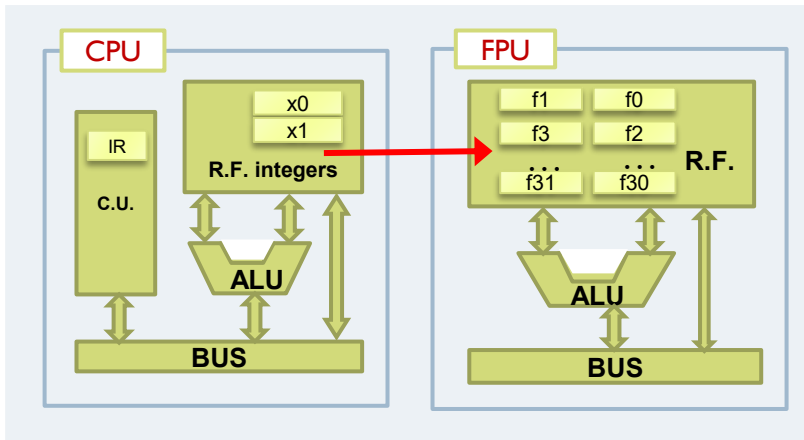
```
main:
```

```
# * It doesn't exist
#  fli.s ft0 3.1415
# * 0x40490E56 is the
#  representation IEEE754
#  in hexadecimal for 3.1415
```

```
li      t0, 0x40490E56
```

```
fmv.w.x ft0, t0
```

```
jr ra
```



Conversion operations (1/3)

double precision $\leftrightarrow$ simple precision

► `fcvt.s.d rd, rs1`

- Convert from double precision (value in floating register `rs`) to simple precision (floating register `rd`).

► `fcvt.d.s rd, rs`

- Convert from simple precision (value in floating register `rs`) to double precision (floating register `rd`).

Conversion operations (2/3)

integer $\leftrightarrow$ simple precision

► `fcvt.w.s rd, rs`

- Convert from simple precision (value in floating register `rs`) to a 32-bit integer 32 **with** sign (integer register `rd`).

► `fcvt.wu.s rd, rs`

- Convert from simple precision (value in floating register `rs`) to a 32-bit integer 32 **without** sign (integer register `rd`).

► `fcvt.s.w rd, rs`

- Convert a 32-bit integer 32 **with** sign (value in integer register `rs`) to a simple precision (floating register `rd`).

► `fcvt.s.wu rd, rs`

- Convert a 32-bit integer 32 **without** sign (value in integer register `rs`) to a simple precision (floating register `rd`).

Conversion operations (3/3)

integer <-> double precision

▶ `fcvt.w.d rd, rs`

- ▶ Convert from double precision (value in floating register `rs`) to a 32-bit integer 32 **with** sign (integer register `rd`).

▶ `fcvt.wu.d rd, rs`

- ▶ Convert from double precision (value in floating register `rs`) to a 32-bit integer 32 **without** sign (integer register `rd`).

▶ `fcvt.d.w rd, rs`

- ▶ Convert a 32-bit integer 32 **with** sign (value in integer register `rs`) to a double precision (floating register `rd`).

▶ `fcvt.d.wu rd, rs1`

- ▶ Convert a 32-bit integer 32 **without** sign (value in integer register `rs`) to a double precision (floating register `rd`).

Example

```
float THREE = 3.0;
```

```
.text
```

```
main:
```

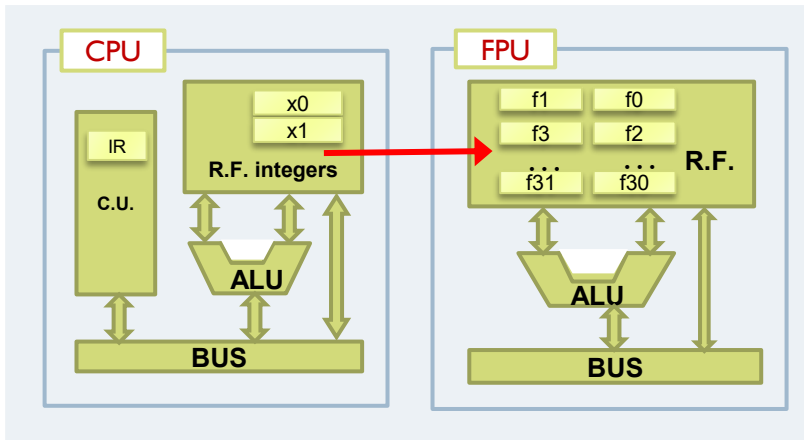
```
# * It doesn't exist
```

```
# fli.s ft0 3.0
```

```
li      t0, 3
```

```
fcvt.s.w ft0, t0
```

```
jr ra
```

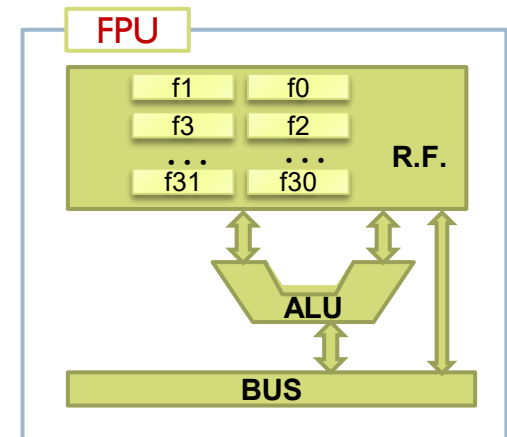


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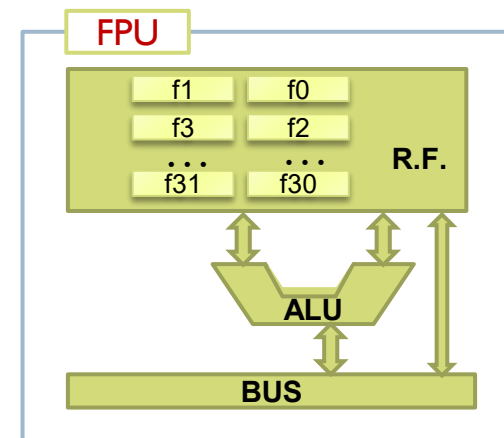
Common arithmetic operations (1/2)

- ▶ Simple (.s) and double (.d) precision:
 - ▶ `fadd.s rd rs1 rs2` $\# \text{rd} = \text{rs1} + \text{rs2}$
 - ▶ `fsub.s rd rs1 rs2` $\# \text{rd} = \text{rs1} - \text{rs2}$
 - ▶ `fmul.s rd rs1 rs2` $\# \text{rd} = \text{rs1} * \text{rs2}$
 - ▶ `fdiv.s rd rs1 rs2` $\# \text{rd} = \text{rs1} / \text{rs2}$
 - ▶ `fabs.s rd rs` $\# \text{rd} = |\text{rs}|$
 - ▶ `fneg.s rd rs` $\# \text{rd} = -\text{rs}$



Common arithmetic operations (2/2)

- ▶ Simple (.s) and double (.d) precision:
 - ▶ `fmin.s rd rs1 rs2` # $rd = \min(rs1, rs2)$
 - ▶ `fmax.s rd rs1 rs2` # $rd = \max(rs1, rs2)$
 - ▶ `fsqrt.s rd rs1` # $rd = \sqrt{rs1}$
 - ▶ `fmadd.s rd rs1 rs2 rs3` # $rd = rs1 \times rs2 + rs3$
 - ▶ `fmsub.s rd rs1 rs2 rs3` # $rd = rs1 \times rs2 - rs3$



Example

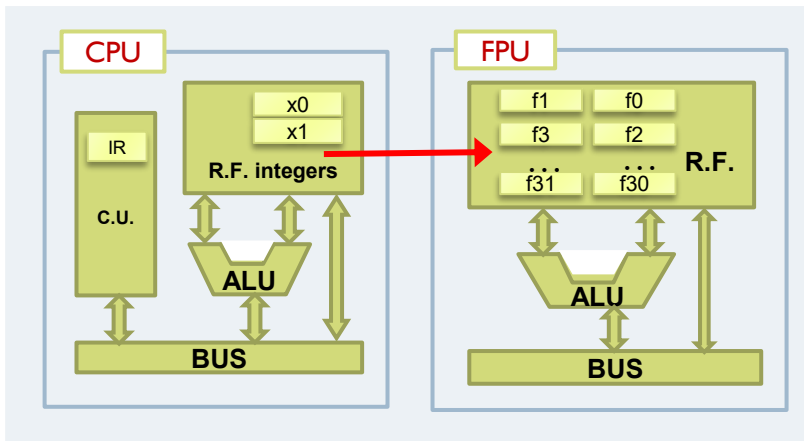
```
float PI    = 3,1415;
int  radio = 4;
float length;

length = PI * radio;
```

```
.text
main:
```

```
li      t0, 0x40490E56
fmv.w.x ft0, t0    # ft0 ← t0
li      t1,  4     # 4 in Ca2
fcvt.s.w ft1, t1    # 4 in ieee754
fmul.s  ft0, ft0, ft1
```

```
jr ra
```



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Classification of floating-point numbers

- ▶ `fclass.s rd, rs1` (simple precision)
- ▶ `fclass.d rd, rs1` (double precision)
- ▶ Save in `rd` the type of the floating-point number of the register `rs1`:

Bit in rd	Meaning
0	-Inf
1	Negative normalized number
2	Negative non-normalized number
3	-0
4	+0
5	Positive non-normalized number
6	Positive normalized number
7	+Inf
8	NaN (non-quiet)
9	NaN (quiet)

IEEE Standard 754 (single precision)

summary

$$(-1)^s \times 0.\text{mantissa} \times 2^{-126}$$

Exponent	Mantissa	Special value
0 (0000 0000)	0	+/- 0 (depends on sign)
0 (0000 0000)	$\neq 0$	Number NOT normalized
255 (1111 1111)	$\neq 0$	NaN (0/0, sqrt(-4),)
255 (1111 1111)	0	+/- infinite (depends on sign)
1-254	Any	Normalized number (no special)

$$(-1)^s \times 1.\text{mantissa} \times 2^{\text{exponent}-127}$$

Example

```
float PI = 3.1415;
```

```
print(typeof(PI));
```

```
.text
```

```
main:
```

```
# ft0 ← 3.0
```

```
li      t0, 3
```

```
fcvt.s.w ft0, t0
```

```
fclass.s t1, ft0
```

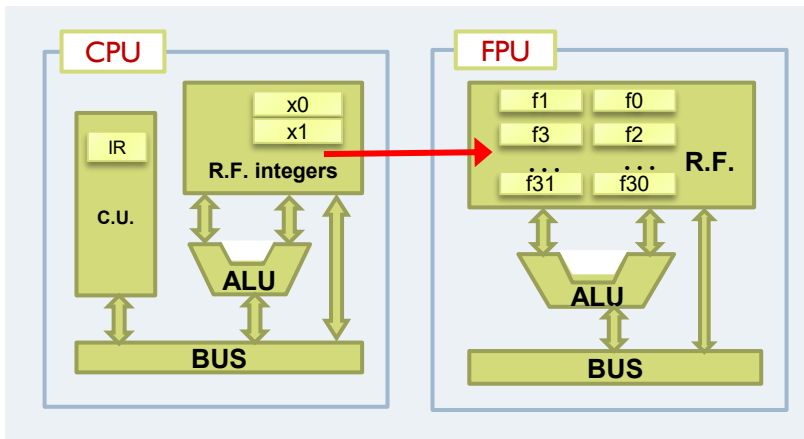
```
# print(typeof(ft0))
```

```
li a7, 1
```

```
mv a0, t1
```

```
ecall
```

```
jr ra
```



Comparison instructions (floating-point registers)

► Simple precision:

- `feq.s rd, rs1, rs2` if (`rs1 == rs2`) `rd=1`; else `rd=0`
- `fle.s rd, rs1, rs2` if (`rs1 <= rs2`) `rd=1`; else `rd=0`
- `flt.s rd, rs1, rs2` if (`rs1 < rs2`) `rd=1`; else `rd=0`

► Doble precision:

- `feq.d rd, rs1, rs2` if (`rs1 == rs2`) `rd=1`; else `rd=0`
- `fle.d rd, rs1, rs2` if (`rs1 <= rs2`) `rd=1`; else `rd=0`
- `flt.d rd, rs1, rs2` if (`rs1 < rs2`) `rd=1`; else `rd=0`

Branches with floating-point numbers

```
x = 3.0;  
y = 2.0;
```

```
if (x < y)  
{  
    . . .  
}
```

```
.text
```

```
main:
```

```
li      t0, 3  
fcvt.s.w ft0, t0
```

```
li      t1, 2  
fcvt.s.w ft1, t1
```

```
flt.s t0, ft1, ft2  
bne t0, x0, end_if1  
# . . .  
end_if1:
```

```
jr ra
```

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Representation of basic data types (1/3)

```
// boolean
bool_t b1 ;
bool_t b2 = false ;
```

```
// character
char c1 ;
char c2 = 'x' ;
```

```
// integers
int res1 ;
int op1 = -10 ;
```

```
// floating point
float f0 ;
float f1 = 1.2 ;
double d2 = 3.0e10 ;
```

```
.data

# boolean
b1:    .zero    1          # 1 bytes
b2:    .byte    0          # 1 byte

# character
c1:    .byte          # 1 bytes
c2:    .byte 'x'      # 1 bytes

# integers
res1:  .zero    4          # 4 bytes
op1:   .word   -10         # 4 bytes

# floating point
f0:    .float          # 4 bytes
f1:    .float    1.2     # 4 bytes
d2:    .double   3.0e10  # 8 bytes
```

Definition of static data (at compile time)

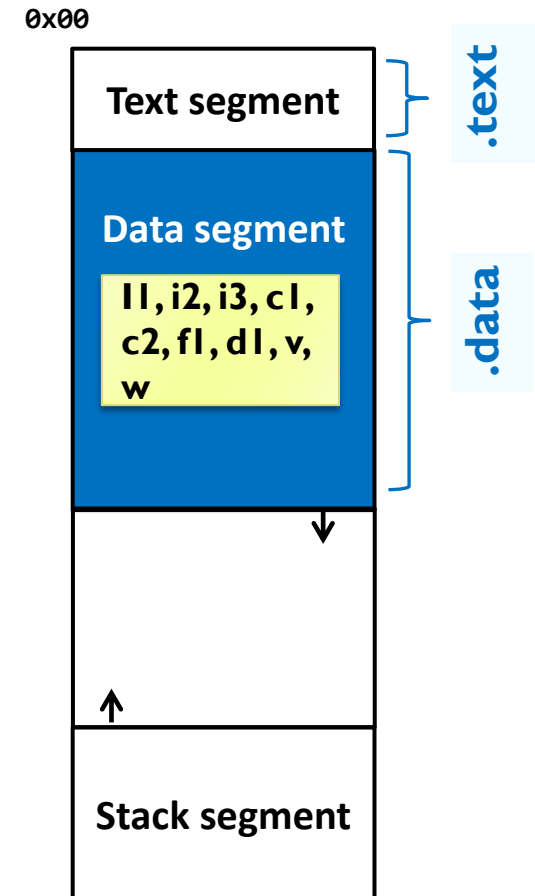
label (address)

Data type (directive)

value

```
.data
cadena: .string "Hello world\n"
i1: .word 10      # int    i1=10
i2: .word -5      # int    i2=-5
i3: .half 300     # short  i3=300
c1: .byte 100     # char   c1=100
c2: .byte 'a'     # char   c2='a'
f1: .float 1.3e-4 # float  f1=1.3e-4
d1: .double .001  # double d1=0.001

# int v[3]={0,-1,0xffffffff}; int w[100];
v: .word 0, -1, 0xffffffff
w: .zero 400
```



Memory operations: floating point

Memory access (floating point)		
<code>flw rs, 10(rs1)</code>	<code>rs = Memory[rs1+10]</code>	Loads the single precision value stored at address (rs1+10) into the floating-point register rd.
<code>fsw rs, 10(rs1)</code>	<code>Memory[rs1+10] = rs</code>	Stores the single value precision of the rs register at address (rd1+10).
<code>fld rd, 10(rs1)</code>	<code>rd = Memory[rs1+10]</code>	Loads the double precision value stored at address (rs1+10) into the rd register.
<code>fsd rd, 10(rs1)</code>	<code>Memory[rs1+10] = rd</code>	Stores the double precision value of the rs register at address (rd1+10).

Elemental data types

Float

```
float  result ;
float  op1 = 100 ;
float  op2 = 2.5
```

...

```
main ()
```

```
{
    result = op1 + op2 ;
    ...
}
```

.data

```
.align 2
result: .zero 4 # 4 bytes
op1:    .float 100
op2:    .float 2.5
# ...
```

.text

```
main:  la      t0,  op1
        flw     ft0, 0(t0)
        la      t0,  op2
        flw     ft1, 0(t0)
        fadd.s  ft3, ft1, ft2
        la      t0,  result
        fsw     ft3, 0(t0)
# ...
```

```
jr ra
```

Elemental data types

Double

```
double result ;
double op1 = 100 ;
double op2 = -10.27 ;
...
```

```
main ()
{
    result = op1 + op2 ;
    ...
}
```

.data

```
.align 2
result: .zero 8 # 8 bytes
op1:    .double 100
op2:    .double 2.5
# ...
```

.text

```
main:  la      t0, op1
        fld     ft0, 0(t0)
        la      t0, op2
        fld     ft1, 0(t0)
        fadd.s  ft3, ft1, ft2
        la      t0, result
        fsd     ft3, 0(t0)
# ...
```

```
jr ra
```

Example. Calculate

$$\sum_{i=1}^N \frac{1}{2^i}$$

```
.data

N:          .word  10           # Upper limit
one_f:      .float 1.0         # 1.0 in floating-point format
two_f:      .float 2.0         # 2.0 in floating-point format

.text

main:
    la      t0, N
    lw      t0, 0(t0)          # t0 = N = 10

    la      t1, one_f
    flw     ft0, 0(t1)         # ft0 = 1.0 (numerator)

    la      t1, two_f
    flw     ft1, 0(t1)         # ft1 = 2.0 (base of the power)

    fmv.w.x fa0, zero          # accumulates the result

    # Initialize the variables for the loop
    li      t1, 1              # t1 = i = 1
    fmv.s   ft2, ft1           # ft2 = 2.0 (will represent 2^i, starting at 2^1)

    # to be continued...
```

Example. Calculate

$$\sum_{i=1}^N \frac{1}{2^i}$$

```
loop:

    bgt t1, t0, end_loop

    # Calculate current term: ft3 = 1.0 / (2^i)
    fdiv.s ft3, ft0, ft2 # ft3 = ft0 / ft2

    # Add to the total accumulator: fa0 = fa0 + ft3
    fadd.s fa0, fa0, ft3

    # Update for the next iteration
    fmul.s ft2, ft2, ft1 # Multiplies the power by 2 for the next power (2^(i+1))
    addi t1, t1, 1      # i = i + 1

j loop

end_loop:

# Print the result
li a7, 2
ecall

li a7, 10    # syscall 10 = exit
ecall
```

Exercise

- ▶ Write a program that:
 - ▶ Loads the value -3.141516 into register f0
 - ▶ Allows you to retrieve the value of the exponent and the mantissa stored in register f0 (in IEEE 754 format)
 - ▶ Prints the sign
 - ▶ Prints the exponent
 - ▶ Prints the mantissa

Exercise (solution)

CREATOR

```
.data
    valor:    .float -3.141516
```

```
.text
```

```
main:
```

```
    la    t0, valor
    flw   ft0, 0(t0)
```

```
    #se imprime
```

```
    fmv.s fa0, ft0
```

```
    li    a7, 2
```

```
    ecall
```

```
    li    a0, '\n'    # enter
```

```
    li    a7, 11
```

```
    ecall
```

```
    # copy to CPU register
```

```
    fmv.x.w    t0,    ft0
```

```
    li    t1, 0x80000000    # sign
    and    a0, t0, t1
    srli    a0, a0, 31
    li    a7, 1
    ecall
```

```
    li    a0, '\n'          # enter
    li    a7, 11
    ecall
```

```
    li    t1, 0x7F800000    # exponent
    and    a0, t0, t1
    srli    a0, a0, 23
    li    a7, 1
    ecall
```

```
    li    a0, '\n'          # enter
    li    a7, 11
    ecall
```

```
    li    t1, 0x007FFFFF    # mantissa
    and    a0, t0, t1
    li    a7, 1
    ecall
```

```
    jr    ra
```

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